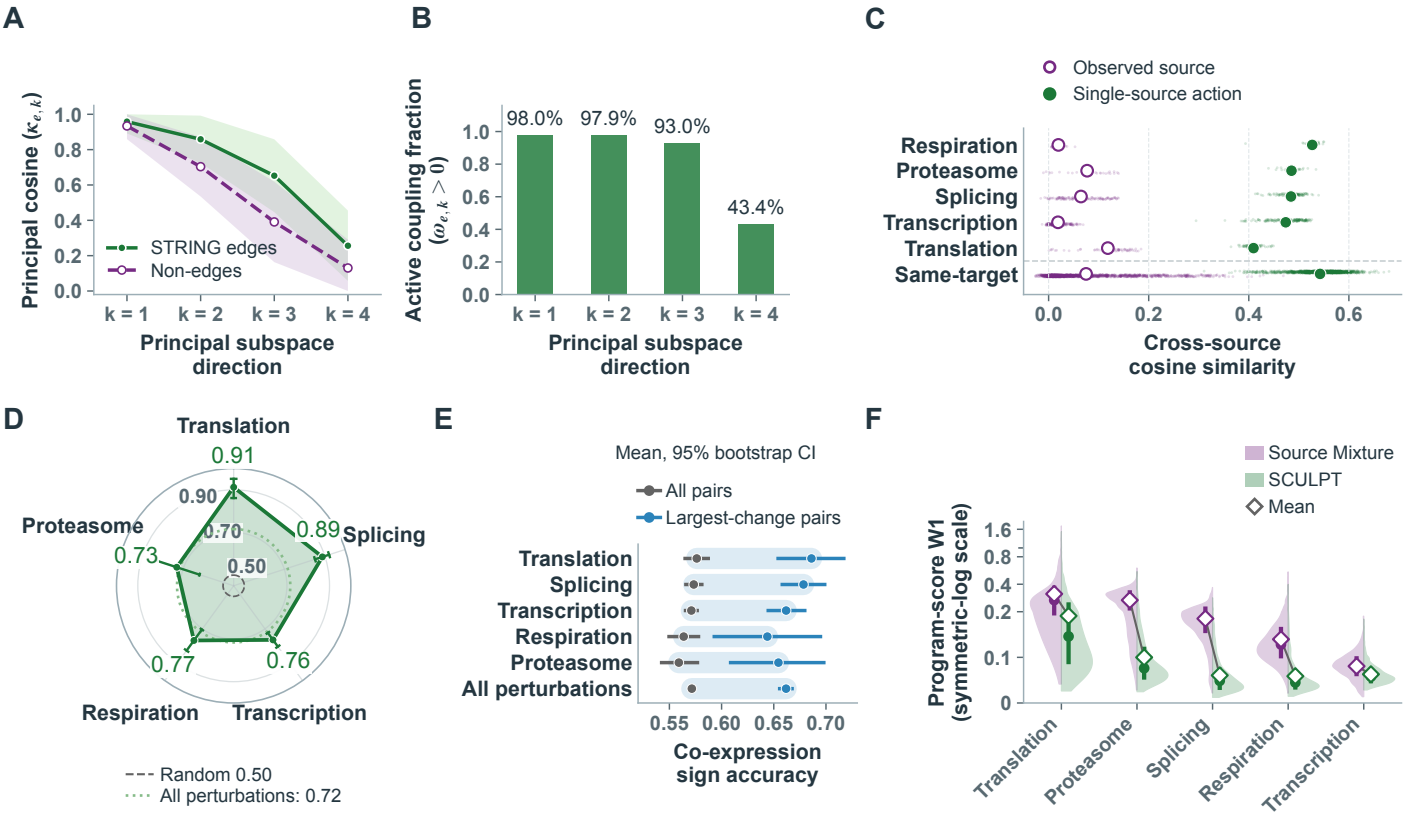

A
Principal cosine ($\kappa_{e,k}$)
1.0
0.8
0.6
0.4
0.2
0.0
STRING edges
Non-edges
k = 1
k = 2
k = 3
k = 4
Principal subspace direction
B
Active coupling fraction ($\omega_{e,k} > 0$)
98.0%
97.9%
93.0%
43.4%
k = 1
k = 2
k = 3
k = 4
Principal subspace direction
C
Observed source
Single-source action
Respiration
Proteasome
Splicing
Transcription
Translation
Same-target
0.0
0.2
0.4
0.6
Cross-source cosine similarity
D
Translation
0.91
0.90
0.70
0.50
Proteasome
0.73
0.89
Splicing
0.77
0.76
Respiration
Transcription
Random 0.50
All perturbations: 0.72
E
Mean, 95% bootstrap CI
All pairs
Largest-change pairs
Translation
Splicing
Transcription
Respiration
Proteasome
All perturbations
0.55
0.60
0.65
0.70
Co-expression sign accuracy
F
Source Mixture
SCULPT
Mean
Program-score W1 (symmetric-log scale)
1.6
0.8
0.4
0.2
0.1
0
Translation
Proteasome
Splicing
Respiration
Transcription